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Master of Computer Applications

Semester – III

Data Analytics for Business

Experiment No: 9

Title: Exploratory data analysis on business datasets

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Problem Statement

You are a Business/Product Analyst tracking the mobile-phone catalog listed on Senior stakeholders want visibility into brand-wise pricing, discounting behaviour, storage/memory configurations, and customer rating quality across the mobile phone portfolio. This experiment creates an interactive e Analytics dashboard that helps evaluate pricing strategy, discount patterns, and product quality across brands, price categories, and storage tiers, using the Dataset.

Github Link: <https://github.com/Amina872/Power-BI>

Methodology

The dataset (Brand, Model, Color, Memory, Storage, Rating, Selling Price, Original Price) was imported into Power BI using Power Query with Unicode (UTF-8) encoding and comma delimiter. New calculated columns (Rating Category, Rating Category Sort, Storage Sort) and DAX measures (Average Selling Price, Average Original Price, Average Rating) were created on the table. All KPI cards, charts, and slicers were then arranged into a single interactive Analytics dashboard. The dashboard includes:

- Total Products KPI Card
- Total Brands KPI Card
- Total Models KPI Card
- Average Selling Price KPI Card
- Average Rating KPI Card
- Average Selling Price by Brand (Bar Chart)
- Original Price vs Selling Price by Brand (Combo Chart)
- Product Rating Distribution (Donut Chart)
- Average Discount % by Brand (Bar Chart)
- Products by Price Category (Column Chart)
- Storage Distribution (Column Chart)
- Interactive Slicers: Brand, Price Category, Memory, Storage, and Rating Category

The dashboard enables management to analyze brand-wise pricing and discounting, monitor product rating quality, and identify which brands, storage tiers, and price segments dominate the catalog, supporting data-driven pricing and inventory decisions.

Data Import

The.csv file was loaded through Power BI's Get Data > Text/CSV connector. File origin was set to 65001: Unicode (UTF-8), the delimiter to Comma, and Data Type Detection to "Based on first 200 rows" before loading the table into the data model.

Flipkart_Mobiles.csv

File Origin: 65001: Unicode (UTF-8) | Delimiter: Comma | Data Type Detection: Based on first 200 rows

Brand	Model	Color	Memory	Storage	Rating	Selling Price	Original Price
OPPO	A53	Moonlight Black	4 GB	64 GB	4.5	11990	15990
OPPO	A53	Mint Cream	4 GB	64 GB	4.5	11990	15990
OPPO	A53	Moonlight Black	6 GB	128 GB	4.3	13990	17990
OPPO	A53	Mint Cream	6 GB	128 GB	4.3	13990	17990
OPPO	A53	Electric Black	4 GB	64 GB	4.5	11990	15990
OPPO	A53	Electric Black	6 GB	128 GB	4.3	13990	17990
OPPO	A12	Deep Blue	4 GB	64 GB	4.4	10490	11990
OPPO	A12	Black	3 GB	32 GB	4.4	9490	10990
OPPO	A12	Blue	3 GB	32 GB	4.4	9490	10990
OPPO	A12	Flowing Silver	3 GB	32 GB	4.4	9490	10990
OPPO	A12	Deep Blue	3 GB	32 GB	4.4	9490	10990
OPPO	A12	Flowing Silver	4 GB	64 GB	4.4	10490	11990
OPPO	A53s 5G	Crystal Blue	6 GB	128 GB	4.3	15990	16990
OPPO	A53s 5G	Ink Black	6 GB	128 GB	4.3	15990	16990
OPPO	A12	Blue	4 GB	64 GB	4.4	10490	11990
OPPO	A53s 5G	Crystal Blue	8 GB	128 GB	4.3	17990	18990
OPPO	A53s 5G	Ink Black	8 GB	128 GB	4.3	17990	18990
OPPO	A33	Moonlight Black	3 GB	32 GB	4.3	10490	12990
OPPO	A31	Lake Green	4 GB	64 GB	4.3	11960	12990
OPPO	A31	Mystery Black	4 GB	64 GB	4.3	11779	11919

The data in the preview has been truncated due to size limits.

Extract Table Using Examples | Load | Transform Data | Cancel

To ensure data quality before loading the table into the data model, two cleaning steps were applied in the Power Query Editor. First, Home > Remove Rows > Remove Duplicates was used across the selected columns to eliminate duplicate product listings from the catalog. Second, the Rating column contained null entries for products with no customer reviews yet; these were standardised by using Transform > Replace Values to replace all null values in the Rating column with 0, so that the column could be used safely in numeric aggregations such as Average Rating.

Power Query Editor interface showing the 'Remove Duplicates' step applied to the 'Flipkart_Mobiles.csv' data source. The 'Remove Duplicates' dialog box is open, indicating that rows containing duplicated values in the currently selected columns will be removed. The data table below shows the resulting list of mobile phones after duplicates have been removed.

Brand	Model	Color	Memory	Storage	Rating	Selling Price	Original Price
OPPO	A53	Mint Cream	4 GB	64 GB	4.5	11990	15990
OPPO	A53	Moonlight Black	4 GB	64 GB	4.5	11990	15990
OPPO	A53	Mint Cream	6 GB	128 GB	4.3	13990	17990
OPPO	A53	Electric Black	4 GB	64 GB	4.5	11990	15990
OPPO	A53	Electric Black	6 GB	128 GB	4.3	13990	17990
OPPO	A12	Deep Blue	4 GB	64 GB	4.4	10490	11990
OPPO	A12	Black	3 GB	32 GB	4.4	9490	10990
OPPO	A12	Blue	3 GB	32 GB	4.4	9490	10990
OPPO	A12	Flowing Silver	3 GB	32 GB	4.4	9490	10990
OPPO	A12	Deep Blue	3 GB	32 GB	4.4	9490	10990
OPPO	A12	Flowing Silver	4 GB	64 GB	4.4	10490	11990
OPPO	A53s 5G	Crystal Blue	6 GB	128 GB	4.3	15990	16990
OPPO	A53s 5G	Ink Black	6 GB	128 GB	4.3	15990	16990
OPPO	A12	Blue	4 GB	64 GB	4.4	10490	11990
OPPO	A53s 5G	Crystal Blue	8 GB	128 GB	4.3	17990	18990
OPPO	A53s 5G	Ink Black	8 GB	128 GB	4.3	17990	18990
OPPO	A33	Moonlight Black	3 GB	32 GB	4.3	10490	12990
OPPO	A31	Lake Green	4 GB	64 GB	4.3	11960	12990
OPPO	A31	Mystery Black	4 GB	64 GB	4.3	11779	11919
OPPO	A74 5G	Fantastic Purple	6 GB	128 GB	4.3	18979	
OPPO	A11K	Flowing Silver	2 GB	32 GB	4.2	8990	
OPPO	F17 Pro	Matte Black	8 GB	128 GB	4.3	19990	
OPPO	A54	Moonlight Gold	4 GB	128 GB	4.3	13990	
OPPO	A54	Moonlight Gold	6 GB	128 GB	4.3	14990	
OPPO	Renov5 5G	Aurora	8 GB	128 GB	4.3	20990	
OPPO	Renov5 5G	Stellar Black	8 GB	128 GB	4.3	20990	
OPPO	F17 Pro	Magic Blue	8 GB	128 GB	4.3	19990	
OPPO	A54	Starry Blue	4 GB	64 GB	4.3	14990	
OPPO	A54	Starry Blue	6 GB	128 GB	4.3	15990	
OPPO	A54	Moonlight Gold	6 GB	128 GB	4.3	16990	
OPPO	A54	Starry Blue	6 GB	128 GB	4.3	16990	
OPPO	A54	Crystal Black	4 GB	64 GB	4.3	14990	
OPPO	F17 Pro	Metallic White	8 GB	128 GB	4.3	19990	
OPPO	A51	Fantasy White	4 GB	64 GB	4.3	12299	
OPPO	F17	New Blue	6 GB	128 GB	4.3	21990	
OPPO	A53	Fancy Blue	6 GB	128 GB	4.3	15999	
OPPO	A16	Pearl Blue	4 GB	64 GB	0	13850	
OPPO	A11K	Deep Blue	2 GB	32 GB	4.2	8990	
OPPO	A54	Crystal Black	4 GB	128 GB	4.3	13990	
OPPO	Renov5 Pro 5G	Aurora Blue	8 GB	128 GB	4.4	30990	
OPPO	A16	CRYSTAL BLACK	4 GB	64 GB	0	13990	

Figure 0a: Removing Duplicate Rows in Power Query

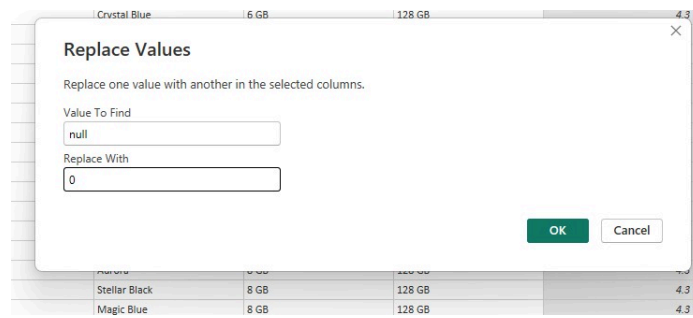


Figure 0b: Replacing Null Values in the Rating Column with 0

Dashboard Images

Image 1st: Overall View (No Filters Applied)

It displays the overall Analytics dashboard without any filters applied (Brand, Price Category, Memory, Storage, and Rating Category slicers set to "All").

- Total Products: 3006 and Total Brands: 17, spanning 888 distinct Models — a large, multi-brand mobile catalog.
- Average Selling Price: ₹25,842 and Average Rating: 4.05, reflecting a catalog skewed toward the budget and mid-range segments but with generally well-rated products.
- The Average Selling Price by Brand chart shows Apple (₹86K) and Google Pixel (₹61K) at the premium end, while Samsung, vivo and ASUS cluster around ₹20K–24K.
- The Original Price vs Selling Price by Brand combo chart shows Apple carrying both the highest original price (₹89K) and selling price (₹62K), the widest brand-level discount gap in absolute terms.
- Product Rating Distribution shows 65.3% of products rated "Very Good (4–4.5)", 17.2% "Excellent (4.5–5)", 12.9% "Good (3–4)", and small slivers for Average and Poor — indicating strong overall product quality.
- Average Discount % by Brand is led by POCO (14.35%) and Motorola (12.54%), while Infixio and realme discount the least (~6.8%–8.7%).
- Products by Price Category peaks at Budget (<₹15K) with ~1.5K products, followed by Mid-Range (~0.8K), High-End (~0.4K) and Premium (~0.3K).
- Storage Distribution is dominated by 64GB (754) and 128GB (752) variants, with 32GB close behind (550); 256GB, 512GB and 1TB together make up a small minority.

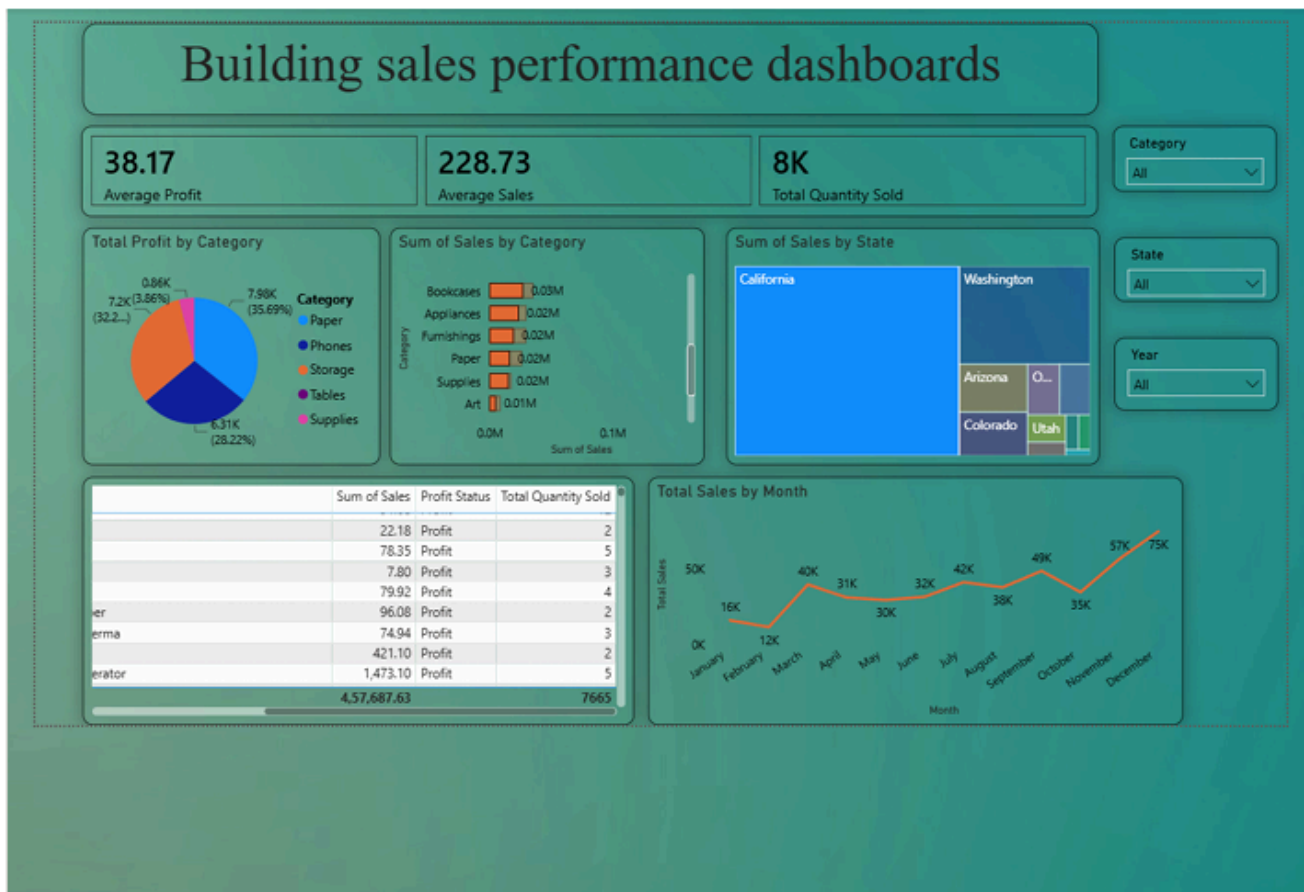


Image 2nd: Filtered by Brand = Apple

In the dashboard, after selecting "Apple" in the Brand slicer:

- All visuals are cross-filtered to show data for the Apple brand only.
- The KPI cards update to Total Products = 324, Total Brands = 1, Total Models = 24, Average Selling Price = ₹85,963, and Average Rating = 3.83.
- The Original Price vs Selling Price by Brand chart collapses to a single Apple bar (₹89K original vs ₹62K selling), confirming Apple is priced well above the portfolio average even after discounting.
- Average Discount % by Brand for Apple is negligible/near-zero compared to brands like POCO and Motorola, showing Apple relies far less on discounting than the rest of the catalog.
- Storage Distribution within Apple skews toward 128GB (71) and 256GB (90), with far fewer 32GB (21) units — consistent with Apple's higher-storage product mix.
- This confirms Apple sits in a distinct premium tier: highest price, lowest discounting, and a rating slightly below the portfolio average (3.83 vs 4.05).

Building sales performance dashboards

38.17

Average Profit

228.73

Average Sales

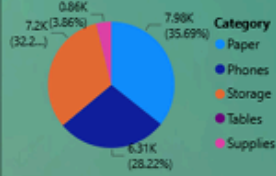
8K

Total Quantity Sold

Category

All

Total Profit by Category



Sum of Sales by Category



Sum of Sales by State



State

All

Year

All

	Sum of Sales	Profit Status	Total Quantity Sold
	22.18	Profit	2
	78.35	Profit	5
	7.80	Profit	3
	79.92	Profit	4
er	96.08	Profit	2
erma	74.94	Profit	3
	421.10	Profit	2
erator	1,473.10	Profit	5
	4,57,687.63		7665

Total Sales by Month



Image 3rd: Filtered by Price Category = Budget (<₹15K)

In the dashboard, after selecting "Budget (<₹15K)" in the Price Category slicer:

- All visuals are cross-filtered to the budget segment only. Total Products drops to 1526 across 15 brands and 560 models.
- Average Selling Price falls sharply to ₹9,452 (vs ₹25,842 overall), as expected for the budget tier, while Average Rating stays healthy at 4.03.
- Average Discount % by Brand remains led by POCO (17.47%) and Motorola (13.26%) even within the budget segment, showing these brands discount aggressively across price tiers, not just at the top end.
- Storage Distribution for budget phones is concentrated in 32GB (412) and 64GB (450), with 128GB (116) and above forming a small share — budget phones are almost entirely lower-storage devices.
- Product Rating Distribution stays similar in shape to the overall view (36.7% Very Good visible slice), showing that budget products are not meaningfully lower-rated than the catalog as a whole.

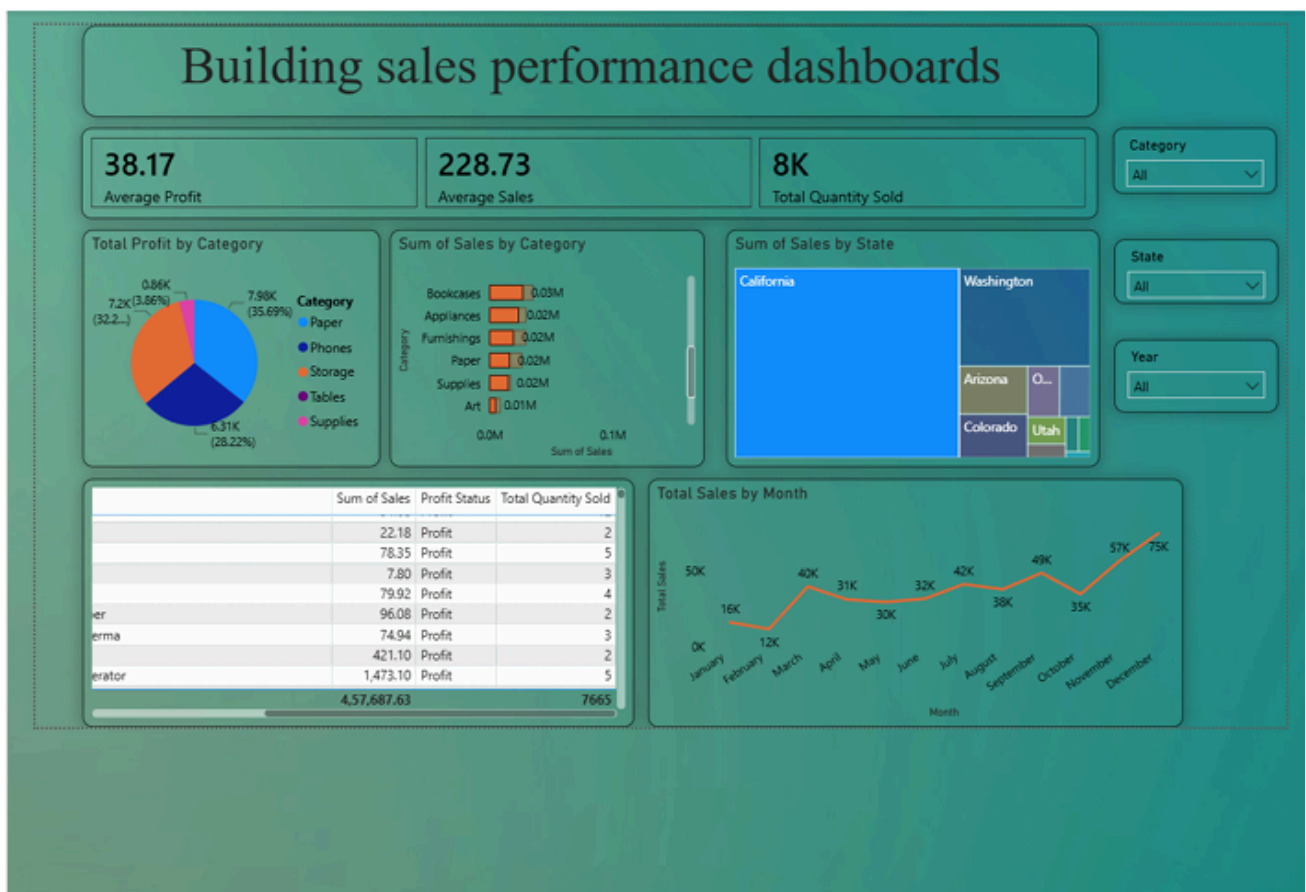
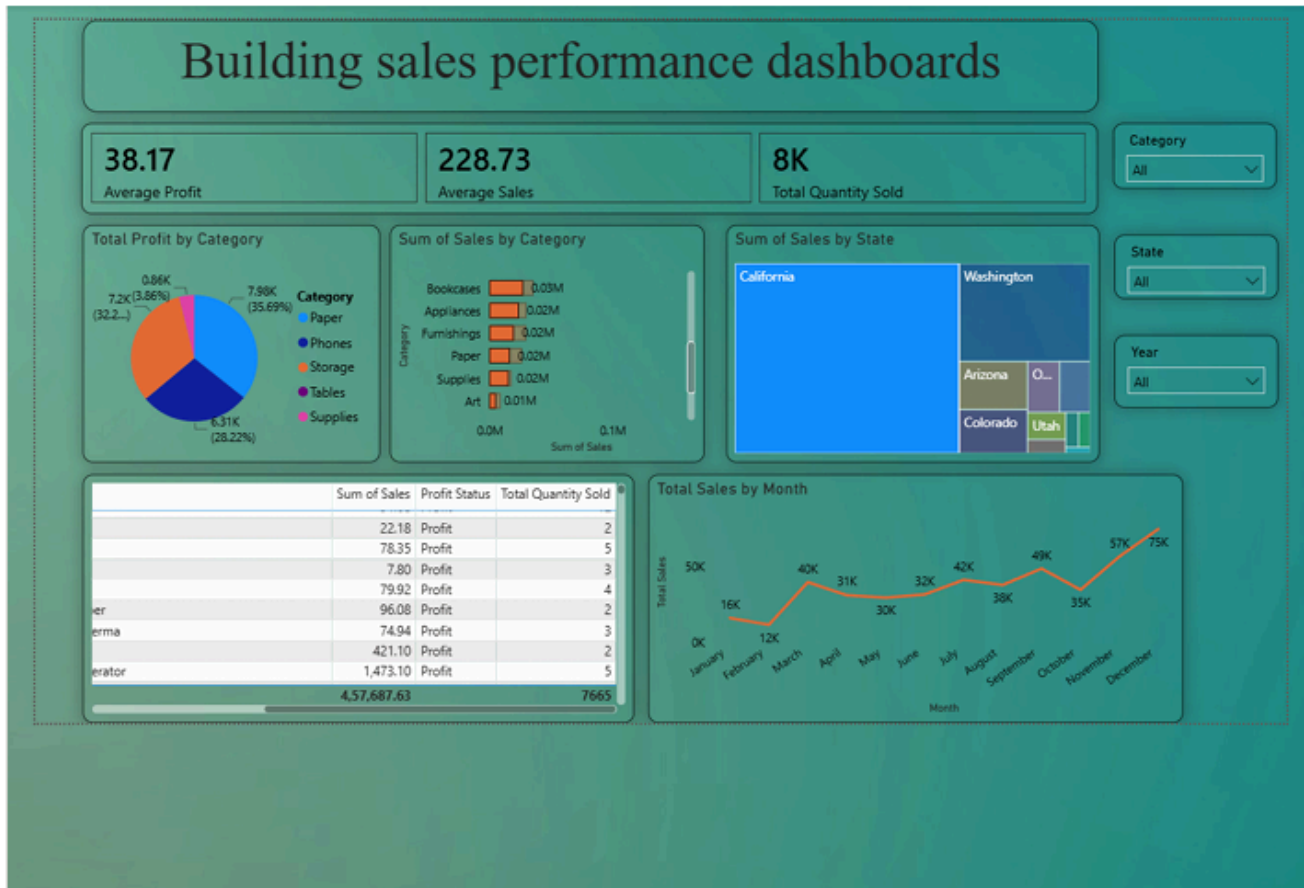


Image 4th: Filtered by Rating Category = Very Good (4–4.5)

In the dashboard, after selecting "Very Good (4–4.5)" in the Rating Category slicer:

- All visuals are cross-filtered to products rated between 4.0 and 4.5. Total Products = 1964 across all 17 brands and 619 models.
- Average Rating naturally narrows to 4.25, and Average Selling Price for this quality band comes out to ₹18,082 — between the overall and budget-only averages.

- Average Discount % by Brand within this rating band is again topped by POCO (16.87%) and Motorola (13.41%), reinforcing that discount depth is a brand-level pattern independent of product rating.
- Storage Distribution for Very-Good-rated phones is fairly balanced across 32GB (431), 64GB (536) and 128GB (521), suggesting well-rated products exist across most storage tiers rather than only at the high end.
- This view helps management identify that the "Very Good" quality band is broad-based across brands and price points, not concentrated in a single segment.



DAX Measures Used

The following DAX measures were created on the table to power the KPI cards and charts in the dashboard:

1. Average Selling Price

```
1 Average Selling Price =  
2 AVERAGE('Flipkart_Mobiles'[Selling Price])
```

Calculates the average selling price across all products. Used to display the Average Selling Price KPI (₹25,842).

2. Average Original Price

```
1 Average Original Price =  
2 AVERAGE('Flipkart_Mobiles'[Original Price])
```

Calculates the average original (MRP) price across all products. Used to power the Original Price vs Selling Price by Brand combo chart.

3. Average Rating

```
1 Average Rating =  
2 AVERAGE('Flipkart_Mobiles'[Rating])
```

Calculates the average customer rating across all products. Used to display the Average Rating KPI (4.05).

New Columns

The following calculated columns were added to the table to support categorisation, sorting, and slicing in the dashboard:

1. Rating Category

```
1 Rating Category =  
2 SWITCH(  
3     TRUE(),  
4     'Flipkart_Mobiles'[Rating] >= 4.5, "Excellent (4.5-5)",  
5     'Flipkart_Mobiles'[Rating] >= 4.0, "Very Good (4-4.5)",  
6     'Flipkart_Mobiles'[Rating] >= 3.0, "Good (3-4)",  
7     'Flipkart_Mobiles'[Rating] >= 2.0, "Average (2-3)",  
8     "Poor (1-2)"  
9 )
```

A SWITCH(TRUE()) column that buckets each product's numeric Rating into five labelled bands: Excellent (4.5–5), Very Good (4–4.5), Good (3–4), Average (2–3), and Poor (1–2). Used to power the Product Rating Distribution donut chart and the Rating Category slicer.

2. Rating Category Sort

```
1 Rating Category Sort =  
2 SWITCH(  
3     TRUE(),  
4     Flipkart_Mobiles[Rating] >= 4.5, 1,  
5     Flipkart_Mobiles[Rating] >= 4.0, 2,  
6     Flipkart_Mobiles[Rating] >= 3.0, 3,  
7     Flipkart_Mobiles[Rating] >= 2.0, 4,  
8     Flipkart_Mobiles[Rating] >= 1.0, 5,  
9     99  
10 )
```

A companion SWITCH(TRUE()) column that assigns a numeric sort key (1–5, with 99 as a fallback) to each Rating Category band, ensuring the donut chart and slicer display bands in a meaningful order (Excellent → Poor) rather than alphabetically.

3. Storage Sort

```
1 Storage Sort =
2 SWITCH(
3     TRUE(),
4     CONTAINSSTRING(UPPER('Flipkart_Mobiles'[Storage]), "32"), 1,
5     CONTAINSSTRING(UPPER('Flipkart_Mobiles'[Storage]), "64"), 2,
6     CONTAINSSTRING(UPPER('Flipkart_Mobiles'[Storage]), "128"), 3,
7     CONTAINSSTRING(UPPER('Flipkart_Mobiles'[Storage]), "256"), 4,
8     CONTAINSSTRING(UPPER('Flipkart_Mobiles'[Storage]), "512"), 5,
9     CONTAINSSTRING(UPPER('Flipkart_Mobiles'[Storage]), "1 TB"), 6,
10    CONTAINSSTRING(UPPER('Flipkart_Mobiles'[Storage]), "1TB"), 6,
11    CONTAINSSTRING(UPPER('Flipkart_Mobiles'[Storage]), "1024"), 6,
12    99
13 )
```

A SWITCH(TRUE()) column that uses CONTAINSSTRING on the Storage text field (32, 64, 128, 256, 512, 1TB/1024) to assign an ascending numeric sort key (1–6, with 99 as a fallback), so the Storage Distribution chart is ordered by capacity (32GB → 1TB) instead of alphabetically.

Business Insights

- The catalog comprises 3,006 products across 17 brands and 888 models, with a healthy Average Rating of 4.05 and 82.5% of products rated "Very Good" or "Excellent" — indicating a generally high-quality listed catalog.
- Apple sits in a clear premium tier of its own: Average Selling Price of ₹85,963 (over 3x the portfolio average), minimal discounting, and only 1 brand/24 models — pricing power comes from brand strength rather than volume.
- POCO and Motorola are the most aggressive discounters (14–17% average discount) across every price segment tested (overall, budget, and Very-Good-rated), suggesting discount-led positioning is a deliberate, consistent strategy for these brands rather than a one-off promotion.
- Budget (<₹15K) phones make up roughly half the catalog (1,526 of 3,006 products) and are concentrated almost entirely in 32GB and 64GB storage, confirming storage capacity is the primary lever brands use to hit budget price points.
- 64GB and 128GB are the dominant storage tiers overall (754 and 752 products respectively), while 256GB+ configurations remain a small niche (under 10% combined) — signalling where most consumer demand and inventory currently sits.
- Product quality (Very Good rating band) is broad-based across brands and storage tiers rather than concentrated in premium models, meaning affordable phones are not being rated meaningfully worse than expensive ones.
- Interactive slicers for Brand, Price Category, Memory, Storage, and Rating Category allow management to drill into specific combinations to identify which brands and configurations are driving the strongest price-to-rating value.

Conclusion

The dataset was successfully imported into Microsoft Power BI and transformed into an interactive Analytics dashboard using KPI cards, a combo chart, a donut chart, and multiple bar/column charts, supported by new calculated columns (Rating Category, Rating Category Sort, Storage Sort) and DAX measures (Average Selling Price, Average Original Price, Average Rating). The dashboard effectively tracks brand-wise pricing, discounting behaviour, storage distribution, and product rating quality across the mobile phone catalog. Overall, the interactive dashboard supports data-driven pricing, discounting, and inventory-planning decisions by enabling management to monitor and evaluate the product portfolio efficiently.

<https://github.com/Amina872/Power-BI>